

- 7) Which of the following is **NOT** a function of the kidney?
- a) Erythropoietin production
 - b) Vitamin D modification
 - c) Acid base balance
 - ☒ d) Aldosterone production
 - e) Renin production
- 8) Which of the following is **NOT** the function of **mesangial cells**?
- a) Structural support to the glomerulus.
 - b) Synthesis of extracellular matrix.
 - c) Endocytosis and contractility.
 - ☒ d) Maintenance of acid-base balance in the blood.
 - e) Increase blood flow and the effective surface area available for filtration.
- 9) Which of the following is **NOT** considered part of the **male genital duct system**?
- a) Rete testis
 - b) Tubuli recti
 - ☒ c) Seminal vesicle
 - d) Ductus deferens
 - e) Ductus epididymidis
- 10) Select the **incorrect** statement in the **male reproductive system**.
- ☒ a) During arousal, the testes normally descend into the scrotum, otherwise they are abdominal organs.
 - b) The bulbourethral glands in males are homologous to the Bartholin's glands in females.
 - c) The prostate gland is made of compound tubuloalveolar glands.
 - d) The glands of the prostate lie in a fibromuscular stroma.
 - e) The prostate gland is the largest accessory structure of the male reproductive system.
- 11) **Testis** has the following coverings **except**
- ☒ a) Tunica adventitia
 - b) Tunica vasculosa
 - c) Tunica albuginea
 - d) Tunica vaginalis
 - e) All of them
- 12) **Fructose**, a potential fuel for mitochondria of spermatozoa, is found in the highest concentrations in secretions of which of the following glands?
- a) Bulbourethral gland
 - b) Glands of Littre
 - c) Prostate gland
 - ☒ d) Seminal vesicle
 - e) Testis
- 13) **Spermiogenesis**.
- a) Occurs before puberty.
 - ☒ b) Involves cytodifferentiation of early spermatids.
 - c) Involves spermatocytogenesis, meiosis and spermatogenesis.
 - d) Takes place in diploid cells.
 - e) Results in the production of primary spermatocytes.
- 14) In the **male reproductive system**, which statement is **false**?
- a) Seminal vesicles are not reservoirs for spermatozoa.
 - ☒ b) 70% of human ejaculate originates in the prostate gland.
 - c) Height of epithelial cells of the seminal vesicle and the degree of activity of the secretory processes are dependant of testosterone levels.
 - d) The tubuloalveolar glands of the bulbourethral glands are formed by simple cuboidal epithelium.
 - e) The number of corpora amylacea increases with age.
- 15) Which statement is **false** about the Sertoli cells?
- a) Their bases adhere to the basal lamina.
 - ☒ b) They produce testicular androgens.
 - c) They may be connected by gap junctions that provide ionic and chemical coupling of the spermatogenic cells.
 - d) They do not divide during the reproductive period.
 - e) They are extremely resistant to adverse conditions such as infections, malnutrition.

7) The labia minora arise embryologically from

- a) Phallus
- b) Labioscrotal swellings
- c) Paramesonephric duct
- d) Urogenital folds
- e) Sinus vaginalis bulb

8) With regard to the development of the male reproductive system, which statement is false?

- a) The genotype of an embryo is established at fertilization.
- b) By week 12, male characteristics of the external genitalia are recognizable.
- c) Sertoli cells are derived from mesoderm.
- d) Seminiferous tubules are formed by week 20.
- e) In epispadias usually the penis is underdeveloped and curved ventrally a condition called chordee.

9) The Y chromosome carries a gene on its short arm that codes for

- a) Testosterone
- b) MIF
- c) Testes-determining factor (TDF)
- d) Progesterone
- e) Estrogen

10) The most common cause of testicular feminization syndrome is

- a) A 46, XO genotype
- b) A 47, XXY genotype
- c) Lack of androgen receptors
- d) Congenital adrenal hyperplasia
- e) Inadequate production of testosterone and müllerian-inhibiting factor (MIF)

11) In the male, failure of urethral folds to fuse completely results in

- a) hypospadias
- b) epispadias
- c) cryptorchidism
- d) hydrocele
- e) congenital inguinal hernia

12) A structure found within the adult female pelvis formed from the gubernaculum is

- a) Broad ligament
- b) Superior ligament of the ovary
- c) Medial umbilical ligament
- d) Median umbilical ligament
- e) Round ligament of the uterus

13) During the development of the female reproductive system, which statement is false?

- a) The phallus forms the mons pubis
- b) The labioscrotal swellings form the labia majora.
- c) Didelphys uterus occurs when there is a complete lack of fusion of paramesonephric ducts.
- d) The urogenital folds remain unfused and form the labia minora.
- e) The cranial portion of the paramesonephric ducts form the uterine tubes.

14) With regard to the development of gonads, which statement is false?

- a) Sex of an embryo is determined genetically at the time of fertilization.
- b) Germ cells do not appear in the genital ridges until the 6th week of development.
- c) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- d) The surface epithelium of the female gonad develop into oogonia.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

15) Concerning the vestigial structures derived from genital ducts, one statement is false?

- a) The cranial end of mesonephric duct in males may persist as paradidymis.
- b) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
- c) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- d) Fused caudal ends of paramesonephric ducts commonly seen in prostate gland as a midline prostatic utricle, which represents the rudimentary uterine primordium.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hyland of Morgagni.

16. Bile canaliculi are lined by:
a) Cholangiocytes
c) Simple columnar epithelium
e) Partly by hepatocytes and partly by cholangiocytes
b) Simple squamous epithelium
d) Hepatocytes
17. Which of the following gland is found on the inner aspect of the lip?
a) Sebaceous gland
b) Sweat gland
c) Mucous gland
d) Apocrine glands
e) All of the above
18. Mucous cells of the salivary glands:
a) Appear translucent after H & E staining
b) Have oval nucleus placed centrally in the cytoplasm
c) Produce thin secretion
d) Usually found as acinar glands
e) Contain zymogen granules in the cytoplasm of the apical aspect
19. Dentin:
a) Is similar to bone, but softer than bone
b) Has many cells located in their lacunae
c) Makes the bulk tissue of the teeth
d) Has capillaries in the dentinal tubules
e) Is heavily innervated
20. Which of the following is **not true** concerning the periodontal membrane?
a) Is made by dense connective tissue
b) Has collagen that live longer, some as same as the teeth
c) Made by bundles of collagen fibres running in different directions
d) May show cementicles (calcified tissue)
e) None of the above
21. Which of the following is **not** found in the appendix
a) Small and irregular lumen
b) Taenia coli
c) Intestinal glands of Lieberkühn
d) Serosa
e) Lymphoid nodules in the lamina propria and submucosa.
22. Which of the following region of the GIT **does not** contain submucosal gland?
a) Esophagus
b) Stomach
c) Duodenum
d) Anal canal
e) All of the above
23. Stellate or Ito's cells found in the liver are located:
a) At the portal canal
b) Space of Disse
c) Space of Moll
d) In the liver sinusoids
e) Around the central vein
24. Portal lobule of liver is demarcated by:
a) Alternate corners at central veins and portal canals of two adjacent classic lobules
b) Portal triad at centre and central veins of adjacent 3 classic liver lobules at the corners
c) Portal triad at centre and central veins of adjacent 4 classic liver lobules at the corners
d) Central vein at the centre and portal triads of adjacent 6 classic liver lobules at the corners
e) Central vein at the centre and 3 portal triads of classic liver lobules at the corners

25. The cells of endocrine portion of pancreas differs from the exocrine portion by being:
 a) Lightly stained b) Found as cords c) Smaller in size
☒ d) All of the above e) None of the above
26. Which type of papillae of the tongue have no taste bud?
 a) Foliate b) Fungiform ☒ c) Filiform
 d) Circumvallate e) None
27. Which of the following cells in the wall of the respiratory tract secrete a lipoprotein surface-active agent that prevents luminal adhesion during expiration:
☒ a) Clara cells b) Goblet cells c) Brush cells d) Basal cells e) Dust cells
28. Which of the following cartilages of the respiratory tract is elastic in type?
 a) Thyroid b) Cricoid ☒ c) Epiglottis
 d) All of the above e) None of the above
29. Which of the following type of cells is **not present** in the respiratory epithelium of the trachea?
 a) Ciliated columnar cells b) Goblet cells c) Basal cells
 d) Stem cells ☒ e) None of the above
30. The olfactory region of nasal cavity
☒ a) The olfactory cells are bipolar neurons
 b) Is made by psuedostatified ciliated epithelium with goblet cells
 c) Has abundant mucous, but no serous producing secretory cells
 d) All are collect e) None of the above
31. Aleveolar epithelium that make the blood air barrier are:
☒ a) Simple squamous b) Simple cuboidal c) Simple columnar
 d) Pseudostratified e) Transitional
32. A renal lobule is made by:
☒ a) A medullary ray and neighboring cortical tissue.
 b) A pyramid and cortex over its base.
 c) A renal column at the side of the pyramid.
 d) A small compartment of the pyramid.
 e) One individual pyramid.
33. Which of the following is the modified smooth muscle cell responsible for the secretion of rennin?
☒ a) Juxtaglomerular cell b) Mesangial cell c) Podocyte
 d) Polkissen (extraglomerular mesangial cell) e) Macula densa cells

34. Which of the following is characteristic of the proximal **but not** of the distal convoluted tubules?
- a) Convoluted portion lies in the renal cortex
 - ☒ b) Exhibits a brush border
 - c) Epithelial lining is similar to that of the loop of Henle
 - d) Lining cells are basophilic
 - e) All of the above
35. The outer layer of Boumans capsule is lined by which type of the epithelium?
- a) Transitional
 - b) Pseudostartified
 - ☒ c) Simple squamous
 - d) Simple cuboidal
 - e) Simple columnar
36. Which of the following region of the urinary tract is lined by staratified columnar epithelium?
- a) Renal calyces
 - b) Renal pelvis
 - ☒ c) Urethra
 - d) Urinary bladder
 - e) Ureter
37. Which one of the following belongs to the acidophils of anterior lobe of pituitary gland?
- ☒ a) Mammothrophs
 - b) Corticotrophs
 - c) Thyrotrophs
 - d) Gonadotrophs
 - e) All
38. Colloids lined by cuboidal cells are found in which region of the hypophysis cerebri?
- a) Pars distalis
 - ☒ b) Pars intermedia
 - c) Pars tuberalis
 - d) Pars media
 - e) Pars nervosa
39. Which endocrine gland contains corpora arenacea?
- a) Adenohypophysis
 - b) Adrenals
 - ☒ c) Pineal body
 - d) Thyroid
 - e) Parathyroid
40. The correct order of the three zones of adrenal cortex outer to inner is:
- ☒ a) Zona glomerulosa, zona fasciculata and zona reticularis
 - b) Zona glomerulosa, zona reticularis and zona fasciculata
 - c) Zona fasciculata, zona reticularis and gona glomerulosa
 - d) Zona fasciculata, gona glomerulosa and zona reticularis
 - e) Zona reticularis, gona glomerulosa and zona fasciculata
41. Mineralocorticoides are produced by cells of which region of the adrenal gland?
- ☒ a) Zona glomerulosa
 - b) Zona reticularis
 - c) Zona fasciculata
 - d) Medulla
 - e) None of the above
42. Chromaffin reaction is a characteristic feature of cells of which region of adrenal gla
- a) Zona glomerulosa
 - b) Zona reticularis
 - c) Zona fasciculata
 - ☒ d) Medulla
 - e) None of the above

43. The blood testis barrier is located between:
- a) Spermatogonia and basal lamina
 - ☒ b) Spermatogonia and primary spermatocytes
 - c) Primary spermatocytes and secondary spermatocytes
 - d) Secondary spermatocytes and spermatids
 - e) Spermatids and spermatozoa
44. During the spermeiogenesis which of the following organelles is primarily involved in the formation of the acrosomal vesicle?
- a) Mitochondria
 - b) Lysosomes
 - ☒ c) Golgi body
 - d) Centrioles
 - e) Smooth endoplasmic reticulum
45. Which of the following male genital duct has a thick muscular wall with a narrow lumen?
- a) Epididymis
 - b) Tubuli recti
 - ☒ c) Ductus deferens
 - d) Ductuli efferentes
 - e) Rete testis
46. Accessory male genital gland that has corpora amylacea or concretions is:
- ☒ a) Prostate gland
 - b) Seminal gland
 - c) Bulbourethral gland
 - d) all of the above
 - e) None of the above
47. What gland in the male is homologous to the greater vestibular gland in the female?
- a) Prostate gland
 - b) Seminal gland
 - ☒ c) Bulbourethral gland
 - d) a & b
 - ☒ e) All
48. Which type of ovarian follicles constitute all the follicles found in the ovary prior to puberty and most found thereafter?
- ☒ a) Primordial follicles
 - b) Primary follicles
 - ☒ c) Secondary follicle
 - d) Antral follicles
 - e) Graafian follicles
49. Which of the following will be excluded from the placental barrier during the 2nd half of pregnancy?
- a) Syncytiotrophoblast
 - ☒ b) Cytotrophoblast
 - c) Basement membrane
 - d) a & b
 - e) None of the above
50. Which layer of the endometrium has lots of decidual cells at the secretory phase of the uterine cycle?
- a) Stratum basalis
 - b) Stratum spongiosum
 - ☒ c) Stratum compactum
 - d) a & b
 - e) All of the above

6. Body of the nail or the nail plate is equivalent to:
 a) Epidermis, dermis and subcutaneous tissue of the skin.
 b) Epidermis and dermis of the skin.
 c) All layers of epidermis with no dermis.
 d) ☒ Stratum corneum of epidermis.
 e) Stratum germinativum of epidermis.
7. The type of epithelium lining the pharynx is:
 a) Simple cuboidal
 b) Simple columnar
 c) Simple squamous
 d) Pseudostaratified
 e) ☒ Stratified squamous nonkeratinized
8. Which of the following is **not** found in the wall of the esophagus?
 a) Lamina propria
 b) Adventitia
 c) Submucosa
 d) All of the above
 e) ☒ None of the above
9. Which of the following regions of the GIT is lined by epithelial secretory sheet?
 a) Esophagus
 b) ☒ Stomach
 c) Small intestine
 d) Large intestine
 e) All of the above
10. The muscularis externa of stomach is made by:
 a) Inner circular and outer longitudinal layer
 b) Inner longitudinal and outer circular layers
 c) ☒ Inner oblique, middle circular and outer longitudinal layers
 d) Inner circular, middle oblique and outer longitudinal layers
 e) Inner longitudinal, middle circular and outer oblique layers
11. Which of the following regions of stomach have similar histology?
 a) Cardiac & Fundus
 b) Cardiac & body
 c) ☒ Fundus and body
 d) Pylorus and cardiac
 e) Body and pylorus
12. During your histology practical session, in which of the following regions of stomach did you see relatively very long gastric glands?
 a) Pylorus
 b) Body
 c) ☒ Fundus
 d) a & b
 e) b & c
13. Peyer's patches of ileum are found in:
 a) Lamina propria
 b) Submucosa
 c) Between inner circular and outer longitudinal muscle layers
 d) ☒ a & b
 e) All of the above
14. The intestinal glands found in the large intestine are:
 a) Compound alveolar
 b) Compound tubular
 c) Compound tubuloacinar
 d) ☒ Simple tubular
 e) Simple acinar
15. The tainea colli of the large intestine are made by:
 a) Circular muscle layer
 b) ☒ Longitudinal muscle layer
 c) Lamina pro
 d) a & c
 e) b & c

10. All the following arteries are found in the abdominal cavity except one?

- a) Superior phrenic arteries
- b) Middle suprarenal arteries
- c) Testicular arteries
- d) Ovarian arteries
- e) None of the above

11. Venous drainage from the head and neck could be by all the following veins except one?

- a) Vertebral veins
- b) Internal jugular veins
- c) Dural sinuses
- d) Subclavian veins
- e) None of the above

12. Which type of cartilage is found in the walls of the eustachian tube?

- A. Elastic cartilage
- B. Hyaline cartilage
- C. Fibrocartilage
- D. All of the above
- E. None of the above

13. Multinucleated cells involved in the resorptions and remodeling of bone tissue

- A. Osteocytes
- B. Osteoblasts
- C. Osteoclasts
- D. Osteoprogenitor cells
- E. Osteon

14. Which type of cartilage forms the epiphyseal growth plate?

- A. Hyaline cartilage
- B. Elastic cartilage
- C. Fibrocartilage
- D. All of the above
- E. None of the above

15. Cell bodies of motor neurons associated with spinal motor nerve roots may be located:

- a. Within the epimysium of the muscle they innervate.
- b. In anterior gray horn of the spinal cord
- c. In dorsal root ganglia.
- d. At any point along a motor nerve.
- e. In the cerebral cortex.

16. In which level of the vertebrae the spinal cord is located by the 6th month of the intrauterine life
- Extends the entire length of embryo
 - At the level of S.1
 - At the level of L2 or L3
 - At the level of lower border of L.1
 - At the level of T12

PART II: GENERAL HISTOLOGY (30 QUESTIONS)

17. Following Haematoxylin & Eosin (H & E) staining, the eosinophilic components of the tissue:
- Stain red-pink
 - Include the nucleus
 - May show metachromatic reaction
 - Stain the ribosomes
 - Are acidic components of the tissue
18. The final magnification of tissues you see under the light microscope may be found:
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 - By the product of magnifying powers of objective lens and eye piece lens.
 - By the product of magnifying powers of condenser lens, objective lens & eye piece lens.
 - Only as the magnifying power of the objective lens.
19. The plasma membrane:
- Has peripheral proteins which are mostly found loosely attached on the outer surface of the membrane.
 - Has integral proteins which show free lateral mobility and rearrangements.
 - Has oligosaccharides which are found projecting from both outer and inner surfaces.
 - Is involved in signal transduction through different classes of polysaccharides receptors.
 - Is clearly visible under light microscopy.
20. Centrioles are constructed from:
- Lysosomes
 - Peroxisomes
 - Microfilaments
 - Microtubules
 - Intermediate filaments
21. Organelles involved in the synthesis of polysaccharides, and glycosylate proteins and lipid and make them ready to be secreted are:
- Rough endoplasmic reticulum
 - Smooth endoplasmic reticulum
 - Peroxisomes
 - Golgi bodies
 - Lysosomes
22. Which of the following organelles are used for intracellular transport system?
- Microtubules
 - Peroxisomes
 - Microfilaments
 - Smooth endoplasmic reticulum.
 - Intermediate filaments

ADDIS ABABA UNIVERSITY
COLLEGE OF HEALTH SCIENCES
SCHOOL OF MEDICINE

MODULE 1 FINAL EXAMINATION FOR FIRST YEAR MEDICAL STUDENTS

Friday, May 16, 2014
Time Allotted: 2:30 hrs

Name: _____ ID Number _____

INSTRUCTION 1: READ THE FOLLOWING QUESTIONS AND CHOOSE THE BEST ANSWER AND WRITE THE LETTER OF YOUR CHOICE ON THE ANSWER SHEET PROVIDED, ATTACHED AT THE LAST PAGE.

PART I: GROSS ANATOMY (16 QUESTIONS)

1. Which of the following vertebral body is heart shaped and contains one or two costal facets?
 - a. Lumbar
 - b. Thoracic
 - c. Cervical
 - d. Sacral
 - e. All of the above
2. Which of the following is **Wrong** about the intervertebral foramen?
 - a. Formed by the vertebral notch of adjacent vertebral pedicles
 - b. An exit for spinal nerves from the spinal canal
 - c. Its posterior border is formed by the zygapophysial joints of the articular processes.
 - d. A herniated intervertebral disc can result narrowing of the intervertebral foramen and may compress the spinal nerve roots.
 - e. None of the above
3. Which of the following is **not** from characteristic features of the extrinsic back muscles?
 - a. They are innervated by the posterior rami of segmental spinal nerves
 - b. They are innervated by the anterior rami of segmental spinal nerves
 - c. The intermediate group muscles have respiratory function
 - d. The superficial group muscles move the upper limb
 - e. None of the above
4. Orientation of the zygapophysial joints of the back show differences that results differences in the degrees of movement in the different regions of the back. Which of the following describes the orientation and movement in the lumbar region?
 - a. Vertical orientation that limits flexion and extension, but facilitate rotation
 - b. Interlock, thereby limiting range of movement
 - c. slope inferiorly from anterior to posterior and facilitate flexion & extension
 - d. Vertical orientation with limiting range of movement
 - e. Horizontal orientation that facilitates rotation

5. Which of the following is **not** muscle of the axial skeleton?
- a) Zygomaticus major muscle
 - b) Brachialis muscle
 - c) Levator ani muscle
 - d) Quadratus lumborum muscle
 - e) None of the above
6. Which of the following is a **wrong** matching between the injury and the affected muscle?
- a. Thoracodorsal nerve injury----- weakness in latissimus dorsi muscle
 - b. Dorsal scapular nerve injury----- weakness in rhomboid minor muscle
 - c. Accessory spinal nerve injury ----- Weakness in rhomboid major muscle
 - d. Injury to cranial nerve eleven (CN XI) ----- Weakness in trapezius muscle
 - e. Dorsal scapular nerve injury ----- Weakness in rhomboids muscles
7. **Wrong** statement about intervertebral disc (IV)?
- a. Posterolateral herniations of the nucleus pulposus are most common in the lumbar and
 - b. Degenerated annulus fibrosus allows herniation of the nucleus pulposus into the vertebral canal
 - c. The outer cartilaginous ring of the intervertebral disc is formed by annulus fibrosus
 - d. A herniated IV disc that compresses the spinal nerve roots results localized pain.
 - e. None of the above
8. All the following muscles are found in the lower limb **except one**
- a) Biceps femoris muscle
 - b) Tibialis posterior muscle
 - c) Sartorius muscle
 - d) Abductor pollicis longus muscle
 - e) None of the above
9. All the following muscles are found in the upper limb **except one**?
- a) Supinator muscle
 - b) Dorsal interosseous muscle
 - c) Extensor digitorum brevis muscle
 - d) Pronator teres muscle
 - e) Palmar interosseous muscle

23. The outer membrane of the nuclear envelop is continuous with
- Microfilaments
 - Peroxisome
 - Rough endoplasmic reticulum
 - Goli bodies
 - Smooth endoplasmic reticulum
24. Regarding to the nucleolus
- is synonymous of the nucleus
 - is the site of mitochondrial synthesis
 - is membrane bound region within the nucleus
 - is made by regions of chromatin contributed by different chromosomes
 - is not visible under the light microscope
25. During a cell cycle,
- Duplication of centrioles begins and complete within the G₁ phase
 - The S phase is the longest and most variable phase
 - RNA and protein synthesis occur during the G₁ phase
 - Synthesis of DNA occurs during the S phase
 - Replication of the chromosomes occurs during G₁ phase
26. An intermediate filament present in the nuclear lamina and used for organization of the nuclear membrane and perinuclear chromatin is
- Vimentin
 - Desmin
 - Glia fibrillary acid protein
 - Cytokeratin
 - Lamins
27. Which of the following is not inclusion?
- Lipid droplets
 - Endosomes
 - Glycogen granules
 - Lipofuscin
 - Melanin
28. An electron microscope is needed to observe all the following features of epithelia except
- Zonula occludens
 - Hemidesmosomes
 - Basement membrane
 - Cilia
 - Flagella
29. Which of the following is a wrong statement about gap junctions?
- Each gap junction could be made up of more than ten members of connexons
 - Each connexon is made up of only six connexons
 - Each connexon-containing channel is made up of pairs of connexons
 - Each connexon protein in a given gap junction could have a different chemical property than its neighbor
 - None of the above

- c) In ordinary connective tissue, the predominant component of ground substance is water.
- d) Fibroblast produces and maintains all extracellular matrix components of connective tissue.
- e) None of the above.
37. Concerning the **connective tissue**, which statement is *false*?
- a) Proteoglycans are responsible for the highly viscous character of the ground substance.
- b) In blood, the ground substance lacks stabilizing macromolecules.
- c) Reticular fibers are particularly abundant in the framework of hematopoietic organs.
- d) Macrophages are also antigen-presenting cells.
- e) None of the above.
38. Which *connective tissue cell type* is functionally related to the *basophils*?
- a) Fibroblast
- b) Mast cell
- c) Pericyte
- d) Plasma cell
- e) Myofibroblast
39. With regard to **cartilage**, which statement is *false*?
- a) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- b) Hyaline cartilage is characterized by an amorphous glassy matrix.
- c) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- d) Fibrocartilage is used for endochondral bone ossification.
- e) Fibrocartilage is present in regions subjected to high tension.
40. **Elastic cartilage**.
- a) is found in the annulus fibrosus and intervertebral disks.
- b) the collagen in its matrix is mainly type II.
- c) is the primary skeletal tissue in the fetus.
- d) has no identifiable perichondrium.
- e) is the most widely distributed cartilage type in the body.
41. With regard to **bone**, which statement is *true*?
- a) Bone formation without a pre-existing cartilage model is called endochondral bone formation.
- b) Interstitial lamellae represent smooth layers of bone on the surfaces of joints.
- c) Compact (dense) bone is characterized by the presence of osteons.
- d) Chondroblasts secrete organic components of bone matrix.
- e) Osteoblasts are involved in bone resorption.
42. Which of the following statements is *correct with* regard to the **bone morphology**?
- a) Primary bone has a higher mineral content than the secondary bone.
- b) Haversian's lacunae are calcified collagen bundles of the periosteum that run in concentric manner.
- c) Osteoclasts have processes that run in the form of canaliculi.
- d) Osteoblasts are derived from blood monocytes.
- e) None of the above.

16. In which level of the vertebrae the spinal cord is located by the 6th month of the intrauterine life
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- Rough endoplasmic reticulum
 - Smooth endoplasmic reticulum
 - Peroxisomes
 - Golgi bodies
 - Lysosomes
22. Which of the following organelles are used for intracellular transport system?
- Microtubules
 - Peroxisomes
 - Microfilaments
 - Smooth endoplasmic reticulum
 - Intermediate filaments

23. The outer membrane of the nuclear envelope is continuous with:
- Microfilaments
 - Peroxisome
 - Rough endoplasmic reticulum
 - Coli bodies
 - Smooth endoplasmic reticulum
24. Regarding to the nucleolus:
- Is synonymous of the nucleus
 - Is the site of mitochondrial synthesis
 - Is membrane bound region within the nucleus
 - Is made by regions of chromatin contributed by different chromosomes
 - Is not visible under the light microscope
25. During a cell cycle:
- Duplication of centrioles begins and complete within the G_1 phase
 - The S phase is the longest and most variable phase
 - RNA and protein synthesis occur during the G_1 phase
 - Synthesis of DNA occurs during the S phase
 - Replication of the chromosomes occurs during G_1 phase
26. An intermediate filament present in the nuclear lamina and used for organization of the nuclear envelope and perinuclear chromatin is:
- Vimentin
 - Desmin
 - Glia fibrillary acid protein
 - Cytokeratin
 - Laminin
27. Which of the following is not inclusion?
- Lipid droplets
 - Endosome
 - Glycogen granules
 - Lipofuscin
 - Melanin
28. An electron microscope is needed to observe all the following features of epithelia except one?
- Zonula occludens
 - Hemidesmosomes
 - Basement membrane
 - Cilia
 - Flagella
29. Which of the following is a wrong statement about gap junctions?
- Each gap junction could be made up of more than two numbers of connexons
 - Each connexon is made up of only six connexins
 - Each connexin-coding channel is made up of pair of connexins
 - Each connexin protein in a given gap junction could have a different chemical property than its neighbor
 - Some of the above

62. Which nucleotide serves as a second messenger in signal transduction pathways?

- A. Cytidine diphosphate
- B. Cyclic guanosine monophosphate
- C. Cyclic adenosine diphosphate
- D. Cyclic uridine diphosphate
- E. Cyclic adenosine monophosphate

63. Which one of the following statements regarding the Watson-Crick DNA model is incorrect?

- A. Polynucleotide chains are anti-parallel with each other
- B. Sugar-phosphate chains occupy the core of the double helix
- C. Polynucleotide chains wind around a central axis
- D. Base pairs are stabilized through hydrogen bonding
- E. The double strands of DNA form a right-hand

PART IV: GENERAL EMBRYOLOGY (19 QUESTIONS)

64. Which of the following is the result of spermatogenesis at meiosis II?

- a) Produces two secondary spermatocytes, haploid single structured chromosomes
- b) Produces two secondary spermatocytes, diploid double structured chromosomes
- c) Produces four spermatids, haploid double structured chromosomes
- d) Produces four spermatids, haploid single structured chromosomes
- e) None of the above

65. Which of the following is not part of spermiogenesis?

- a) Condensation of the nucleus and acrosome formation
- b) Shedding of most of the cytoplasm
- c) Entering a prolonged prophase of 22 days
- d) Formation of neck, middle piece and tail
- e) None of the above

66. The _____ is a derivation of cumulus oophorus formed following ovulation of the oocyte.

- a) Corona radiata
- b) Zona pellucid
- c) Theca interna
- d) Polar body
- e) Theca externa

67. Which of the following is not related with a surge of luteinizing hormone during the midovarian cycle?

- a) Causes follicular rupture and ovulation
- b) Stimulates production of progesterone by follicular stromal cells
- c) Causes thinning of cervical mucus
- d) Elevates concentrations of maturation promoting factor
- e) None of the above

68. Which of the following structure among the given alternative comes last developmentally

- A. Epiblast
- B. Secondary yolk sac
- C. Chorionic cavity
- D. Amnion
- E. Prechordal plate

69. Extraembryonic mesoderm is derived from cell arising from

- A. Amnion
- B. Heuser's membrane
- C. Epiblast
- D. Hypoblast
- E. Cytotrophoblast

70. Which of the following is not the function of notochord
- Defines the primordial longitudinal axis of the embryo
 - Provides signals that are necessary for the development of axial musculoskeletal
 - Degenerates as the bodies of the vertebrae form, but small portions of it persist as the *Anulus fibrosus* of each intervertebral disc.
 - Induces the overlying embryonic ectoderm to thicken and form the neural plate
 - None
71. Which of the following tissue is derived from the surface ectoderm
- Anterior lobe of the pituitary gland
 - Retina of the eye
 - The pia and arachnoid mater
 - Multipolar motor neuron
 - The neurons of the sensory ganglia of the cranial nerves
72. The main event which takes place by the seventh week of the developing individual is/are
- Scalp vascular plexus
 - Ossification of bone in the upper limb has begin
 - Appearance of auricular hillocks
 - Formation of Mesonephric ridges
 - Appearance of the fourth pharyngeal arch
73. Which of the following is appropriate diagnostic techniques for accessing the fetal chromosomal abnormalities, inborn errors of metabolism, and X-linked disorders
- Spectrophotometric Studies
 - Chorionic Villus Sampling
 - Sex Chromatin Patterns
 - Cell Cultures and Chromosomal Analysis
 - Percutaneous Umbilical Cord Blood Sampling
74. All are events occurring during the pre embryonic period of the prenatal development except:
- Formation of neural tube.
 - Formation of heart tube from the cardiogenic mesoder.
 - Formation of stomodium and proctodium.
 - Formation tertiary chorionic villa.
 - None of the above
75. Concerning the fetal membranes, which statement is true?
- The yolk sac helps control the embryo's body temperature.
 - As the bladder enlarges, the umbilical cord involutes to form urachus.
 - Blood formation occurs in the wall of allantois during the third to fifth weeks.
 - Primordial germ cells appear in the endodermal lining of the wall of the allantois.
 - During the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.

43. Which of the following cells *does not* belong to the *mononucleated phagocytic system*?
- a) Microglia
 - b) Hepatocytes
 - c) Dust cells
 - d) Langerhan's cells
 - e) Kupffer cells

44. With regard to the *white blood cells*, which of the following statements is *true*?
- a) Neutrophils are the most common white blood cells in circulating blood.
 - b) A patient with a parasitic infection is likely to reveal an increase in the circulating monocytes.
 - c) Basophils normally re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues.
 - d) A macrophage can originate from a mast cell.
 - e) None of the above.
45. Fibrocartilage is not found in the:
- a) Intervertebral discs
 - b) Pubic symphysis
 - c) Discs of knee joint
 - d) Temporomandibular joint
 - e) Epiglottis
46. Select the *incorrect statement* about *mature erythrocytes*.
- a) do not contain nuclei.
 - b) possess haemoglobin, which fills almost the entire cytoplasm.
 - c) are unable to move actively, but they are remarkably elastic and can withstand deformation.
 - d) contain the organelle peroxisome as oxygen binding protein.
 - e) None of the above.

PART III: Biochemistry (17 questions)

47. Lateral diffusion of membrane lipids occurs at a very slow rate.
- a) True
 - b) False
48. Phosphatidic acid is a common structural component of different phospholipids.
- a) True
 - b) False
49. A membrane rich in saturated fatty acids is more fluid than the one rich in unsaturated fatty acids.
- a) True
 - b) False
50. Sucrose and lactose are examples of reducing disaccharides.
- a) True
 - b) False
51. Hyaluronic acid serves as lubricant and shock-absorber in synovial fluids.
- a) True
 - b) False
52. Naturally occurring fatty acids usually contain *trans* double bonds.
- a) True
 - b) False
53. Fats and oils are acylglycerols that differ in the nature of their fatty acids.
- a) True
 - b) False
54. N-acetylglucosamine is a common component of ABO blood group determinants.
- a) True
 - b) False

55. Which type of amino acid serves as a precursor for the biosynthesis of niacin (vitamin B3) in our body?

- A. Glycine
- C. Tryptophan
- D. Cysteine
- E. Lysine

56. Cysteine and methionine

- A. Contain sulfur
- B. Have polar side chains
- C. Have non-polar side chains
- D. Contain imino group
- E. A & C

57. A tripeptide that protects cell membranes from damage with its antioxidant properties is

- A. Corticotropin
- B. Vasopressin
- C. Bradykinin
- D. Angiotensin
- E. Glutathione

58. Collagen contains hydroxylated derivatives of

- A. Arginine and glutamate
- B. Cysteine and tryptophan
- C. Lysine and proline
- D. Tyrosine and dopamine
- E. Valine and isoleucine

59. One of the following proteins differs in architecture and solubility from the others

- A. Immunoglobulin
- B. Keratin
- C. Elastin
- D. Myosin
- E. Fibrinogen

60. Which one of the following could not be found as a component of nucleic acids?

- A. Uracil
- B. Hypoxanthine
- C. Uric acid
- D. Xanthine
- E. B and D

61. The important functional groups in purines and pyrimidines for the formation of base pairs include

- A. The amino groups of guanine and cytosine
- B. The oxygen linked to the fourth carbon of uracil and thymine
- C. The third nitrogen of pyrimidine bases
- D. The first nitrogen of purine bases
- E. All of the above

91. Heterostasis is characterized by:
- Hypervascularization
 - Hyperventilation
 - Tachycardia
 - ☒ All of the above
 - None of the above
92. Adaptation to cold environment is characterized by all of the following except:
- Reduced basal metabolic rate
 - Increased thermal conductivity
 - A skin temperature of 12-15 °C
 - Increased subcutaneous fat
 - Decrease in surface area of the body
93. Which of the following is an example of behavioral response:
- Regulation of arterial blood pressure
 - Regulation of body temperature
 - Regulation of fluid volume
 - Control of diet
 - Release of catecholamines
94. A physiological error signal is:
- The same as a reference value
 - Transmitted to a detector via efferent nerves
 - Sent to effector organs via afferent nerve
 - ☒ The difference between a reference value (set point) and a new value signaled to an effector organ
 - None of the above
95. The rennin-angiotensin-aldosterone system (RAAS) is required in the regulation of:
- Arterial blood pressure
 - Blood volume
 - Fluid osmolality
 - H⁺ and K⁺ secretion
 - All of the above
96. In the establishment of the mechanisms of physiological adjustment Hans Selye:
- Explained the scientific study of stress
 - Coined the term "homeostasis"
 - Stated that homeostasis favours stability
 - Described "illhealth vs. imbalance"
 - Described the relationship between the Autonomic nervous system and fight flight reactions.
97. Multiplicity of receptors and effectors:
- Ensures stability of regulation and survival
 - Occurs in the cardiovascular system
 - Contributes to continuity
 - All of the above are true
 - Leads to instability of internal environment

98. Allostasis

- a. Is attaining viability through acute change
- b. Is adjustment to new conditions through chronic change
- c. When overloaded, may result in excessive stress hormone exposure
- d. May end up with atherosclerosis when elevation of arterial BP is persistently elevated
- e. May be characterized by all of the above except "b".

Select A. if 1,2,3 are correct
 B. if 1,3 are correct
 C if 2 and 4 are correct
 D if only 4 is correct
 E if all four are correct

99. Electrolyte excess/is characterized by:

- 1. Headache and vomiting
- 2. Watery diarrhea
- 3. Excessive drinking
- 4. Thirst

100. Behavioral responses may be exemplified by:

- 1. Regulation of blood glucose
- 2. Regulation of arterial blood pressure
- 3. Regulation of body temperature
- 4. Control of environmental condition

101. Match:

- | | |
|--------------------------------|-------------------------------------|
| 1. Control of diabetes by diet | a) Indicator of allostatic overload |
| 2. Control of body temperature | b) Behavioural response |
| 3. Positive feedback | c) Replication of micro-organisms |
| 4. Negative feedback | d) Reflex mechanism |
| 5. Heterostasis | e) A feedback factor of 5 |
| 6. Metabolic deterioration | f) Hypervascularization |

T/F

102. In the ontogeny of regulatory mechanisms:

- a. Survival precedes
- b. Regulation in time and amount is the last process
- c. Linked functional processes come second
- d. Structural specialization comes first

103. The regulation of the level of glucose in the blood involves

- a. The hypothalamus as a coordinating centre
- b. The hypothalamus as an effector organ
- c. The Liver as a receptor organ
- d. Endocrine pancreas as a receptor organ

76. Concerning **multiple pregnancies**, which statement is **false**?
- The frequency of monozygotic twinning shows marked racial differences.
 - The rate of dizygotic twinning increases with maternal age.
 - Dizygotic twins always have two amnions and two chorions.
 - Dizygotic twinning shows a hereditary tendency.
 - None of the above.
77. **Diethylstilbestrol (DES)** was once prescribed to women to prevent miscarriages. However, it has since been found to be a teratogen that causes what type of defects?
- Reproductive system
 - Central nervous system
 - Gastrointestinal
 - Limb (partial or total absence)
 - Heart and neural tube defects
78. Most **birth defects** occur during:
- The first 2 weeks
 - The embryonic period
 - The fetal period
 - Early childhood
 - None of the above
79. With regard to **birth defects**, which statement is **false**?
- Clubfoot, which is due to mechanical forces in the womb, is an example of deformation.
 - Trisomy of the autosomes occurs with increasing frequency as maternal age increases.
 - Deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
 - The phenotype of Turner syndrome is female.
 - Thalidomide induces limb defects.
80. Which of the following is **not the function of the amniotic fluid**?
- Acts as a barrier to infection
 - Permits normal fetal lung development
 - Prevents adherence of the amnion to the embryo and fetus
 - Cushions the embryo and fetus against injuries by distributing impacts the mother receives
 - None of the above.
81. The pericardioperitoneal canals develop into:
- Pericardial cavity
 - Peritoneal cavity
 - Pleural cavities
 - a & b
 - All of the above
82. With regard to the development of diaphragm:
- The central tendon of the diaphragm develops from dorsal mesentery of the esophagus.
 - Septum transversum gives rise to the cupula of the diaphragm.
 - The costal margin of the diaphragm develops for the pleuropericardial folds.
 - Diaphragm ascends during development.
 - Eventration of the diaphragm could lead to hypoplasia of the lung.

PART V: PHYSIOLOGY (21 QUESTIONS)

83. Symport may be:
- Na⁺- glucose transport
 - Cl⁻-HCO₃ transport
 - Ca⁺⁺-Na⁺ transport
 - Na⁺-K⁺ transport
 - None of the above

84. The cell is characterized by
- Self-preservation
 - Reproduction
 - A 10-20% protein mass
 - The presence of sex-chromatin in its nucleus
 - All of the above
85. Which of the following is the fattest individual?
- An infant with body water content of 65%
 - An infant with body water content of 80%
 - An adult with body water content of 45%
 - An adult with body water content of 60%
 - None of the above
86. A drop in interstitial fluid (IF) volume leads to:
- Low BP
 - Poor skin turgor
 - Dry tongue
 - Rapid pulse
 - (b) and (c) above
87. Which of the following is indicative of heavy exercise?
- A urine output of 500 ml/day
 - A urine output of 1200ml/day
 - A sweat output of 1400ml/day
 - A sweat output of 5L/day
 - "a" and "d" above
88. Causes of pitting oedema include all of the following except:
- Increased hydrostatic pressure in capillaries secondary to heart failure
 - Decreased osmotic pressure due to low plasma proteins secondary to conditions like nephrotic syndrome
 - Increased capillary permeability.
 - Depression of cellular metabolism
 - Localized Increased venous pressure
89. All of the following **except one** are examples of positive feedback mechanism
- Progress of labour
 - Nerve impulse
 - Release of cortisol
 - Replication of viruses
 - Irreversible shock
90. One of the following serves as a detector in the regulation of BP:
- Vasomotor centre
 - Arteriole
 - Aortic body
 - Carotid artery
 - Aortic arch

3. A high-velocity bullet penetrated the posterior aspect of the shoulder and severely injured the origin of the posterior cord of the brachial plexus. Each of the following nerves might possibly be paralyzed, EXCEPT the:
- a) upper subscapular
 - b) axillary
 - c) thoracodorsal
 - d) radial
 - e) medial pectoral
4. As a result of an injury to the ulnar nerve in the arm, all of the following changes in the hand will be noted, EXCEPT:
- a) claw hand
 - b) the thumb is strongly adducted
 - c) loss of flexion of the distal interphalangeal joints of the fourth and fifth digits.
 - d) There is sensory loss on the little finger and one-half of the ring finger
 - e) There is marked wasting in the hypothenar eminence
5. Sensory loss over the skin covering the dorsal surface of the thumb would indicate damage to which of the following nerves?
- a) deep radial
 - b) superficial radial
 - c) anterior interosseous
 - d) median
 - e) ulnar
6. Compression of the median nerve in the carpal tunnel deep to the flexor retinaculum could affect the functioning of each of the following muscles, EXCEPT the:
- a) second lumbrical
 - b) opponens pollicis
 - c) flexor pollicis brevis
 - d) adductor pollicis

7. Which one of the following arteries is not a part of the arterial-anastomosis around the scapula:
- a) thoracodorsal
 - b) subscapular
 - c) suprascapular
 - d) circumflex scapular
 - e) transverse cervical
8. Which one of the following is not true of the breast (mammary-gland):
- a) it lies on pectoralis major & serratus anterior muscles
 - b) cancer of the breast could spread by venous channels
 - c) cancer of the breast in the lower-inner quadrant is most dangerous
 - d) it is supplied by both internal and lateral thoracic arteries
 - e) its medial quadrant is drained by axillary lymph nodes
9. In the cubital fossa, the median cubital vein is frequently used for venipuncture. The vein is separated from underlying neurovascular structure by the:
- A. investing layer of brachialis muscle
 - B. intermuscular septum
 - C. bicipital aponeurosis
 - D. anconeus muscle
 - E. tendon of brachialis muscle
10. A 52-year-old man complained of severe pain in the left arm after a fall. Radiological examination confirmed a broken humerus. The patient was able to extend the forearm at the elbow, but supination appeared somewhat weak; Neurologic examination of a patient revealed an inability to extend the wrist. The observations that extension at the elbow appeared normal, but supination of the forearm weak, indicates localization of the nerve injury to the:
- A. posterior division of the brachial plexus
 - B. posterior cord of the brachial plexus in the axilla
 - C. radial nerve at the distal third of the humerus
 - D. radial nerve injury near the head of the radius
 - E. radial nerve in the mid-forearm

11. If a tumor grows into the quadrangular space of the axillary region, which structures would be in danger?

- A. anterior humeral circumflex artery and axillary nerve
- B. posterior humeral circumflex artery and radial nerve
- C. posterior humeral circumflex artery and axillary nerve
- D. radial nerve and profunda brachii artery
- E. profunda brachii artery and axillary nerve

12. If the entire greater tubercle of the humerus was torn away as a result of an injury, which of the following movements of the humerus would be affected?

- a) flexion and abduction
- b) abduction and lateral rotation
- c) flexion and medial rotation
- d) extension and medial rotation
- e) flexion, abduction, and adduction

13. The superficial palmar arterial arch of the hand:

- a) is located in the thenar compartment
- b) is proximal to the deep palmar arterial arch
- c) is deep to the tendons of the flexor digitorum profundus
- d) is formed primarily by the ulnar artery
- e) courses with the deep branch of the ulnar nerve

14. Each of the following muscles connects the upper limb to the thoracic wall, EXCEPT the:

- a) pectoralis major
- b) pectoralis minor
- c) subclavius
- d) latissimus dorsi
- e) serratus anterior

15. Which of the following nerves sends branches to the muscles of the arm?

- A. median
- B. ulnar
- C. dorsal scapular
- D. musculocutaneous
- E. long thoracic

16. Following a stab wound in the wrist area, a patient is unable to flex the distal phalanx of the middle finger. This would be a result of an injury to the:

- A. superficial ulnar nerve
- B. deep ulnar nerve
- C. median nerve
- D. posterior interosseous nerve
- E. radial nerve

17. The nerve most likely to be injured in fractures of the medial epicondyle is the:

- A. radial
- B. axillary
- C. ulnar
- D. median
- E. musculocutaneous

18. The humerus is a long bone that is closely associated with nerves. Which nerve is injured in fractures of the surgical neck of the humerus?

- a. radial
- b. ulnar
- c. median
- d. musculocutaneous
- e. axillary

19. After radical surgery of the breast for breast cancer, the patient was not able to abduct the arm above the horizontal level. What nerve is paralyzed?

- a. suprascapular
- b. axillary
- c. long thoracic
- d. dorsal scapular
- e. subscapular

20. Which of the following is affected in Dupuytren's Contracture of palmar aponeurosis?

- a. Index finger
- b. Middle finger
- c. Ring finger
- d. Little finger
- e. c and d
- f. a and b

21. Which of the following nerves may be injured in Lower Lesion of Brachial Plexus (Klumpke's Paralysis)?

- a. suprascapular
- b. axillary
- c. ulnar
- d. dorsal scapular
- e. subscapular

22. Which of the following nerves arise directly from the roots of the brachial plexus?

- a. suprascapular
- b. axillary
- c. ulnar
- d. dorsal scapular
- e. subscapular

23. Which of the following arteries arise from the second part of the axillary artery.

- a. superior thoracic
- b. lateral thoracic
- c. thoracodorsal
- d. subscapular
- e. dorsalscapular

24. The thumb action that is **totally** affected by radial nerve trauma is:

- A. abduction
- B. adduction
- C. extension
- D. flexion
- E. opposition

25. The extensor expansion of the fingers is formed by the tendon of the:

- A. extensor carpi radialis longus
- B. extensor pollicis longus
- C. extensor digitorum
- D. extensor pollicis brevis
- E. interossei

26. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this action, which nerve is suspected to be nonfunctional?

- A. radial
- B. median recurrent
- C. ulnar
- D. deep radial
- E. superficial radial

27. Which nerve is probably damaged if a patient cannot abduct the arm beyond 25 degrees?

- A. axillary
- B. radial
- C. musculocutaneous
- D. median
- E. ulnar

28. Loss of pronation of the hand suggests a lesion of the:

- A. median nerve
- B. radial nerve
- C. ulnar nerve
- D. intercostobrachial nerve
- E. musculocutaneous nerve

29. With an injury to the right upper trunk of the brachial plexus (Erb Duchenne paralysis), the patient's right upper extremity would be:

- A. abducted
- B. elevated
- C. medially rotated
- D. laterally rotated
- E. flexed

30. Loss of opposition of the thumb is a symptom associated with lesion of which nerve?

- A. radial
- B. ulnar
- C. musculocutaneous
- D. median
- E. posterior interosseous

31. All of the following muscles attach to the medial border of the scapula, EXCEPT the:

- A. levator scapulae
- B. rhomboideus major
- C. teres major
- D. serratus anterior
- E. rhomboideus minor

32. Which nerve-artery pair do not run together?

- A. radial nerve, profundus brachii artery
- B. axillary nerve, posterior humeral circumflex artery
- C. median nerve, posterior interosseous artery
- D. anterior interosseous nerve, anterior interosseous artery
- E. none of the above

33. Which group of muscles below is supplied by the obturator nerve?

- A. the chief adductors of the thigh
- B. the chief lateral rotators of the thigh
- C. the chief flexors of the thigh
- D. the chief extensors of the thigh
- E. the chief abductors of the thigh

34. A patient presents with sensory loss on adjacent sides of the great and second toes and can not dorsiflex the foot. These signs probably indicate damage to the:

- a) Superficial peroneal nerve
- b) Lateral plantar nerve
- c) Deep peroneal nerve
- d) Sural nerve
- e) Tibial nerve

35. All of the following bones are associated with the medial longitudinal arch of the Foot, EXCEPT:

- a) Talus
- b) Calcaneus
- c) Medial three metatarsal bones
- d) Navicular
- e) Cuboid

36. Dislocation of the hip results in a vascular necrosis of the hip joint due to injuries to all of the following arteries, EXCEPT:

- a) Medial femoral circumflex artery
- b) Inferior gluteal artery
- c) Superior gluteal artery
- d) Lateral femoral circumflex artery
- e) Deep iliac circumflex artery

37. Nerves can frequently be compressed against bony structures in the lower limb. The nerve that rests against the head and neck of the fibula is the:

- a) Saphenous
- b) Common peroneal
- c) Lateral sural cutaneous
- d) Tibial
- e) Medial sural cutaneous

38. Which group of muscles below is supplied by the femoral nerve?

- a. the chief adductors of the thigh
- b. the chief lateral rotators of the thigh
- c. the chief abductors of the thigh
- d. the chief flexors of the leg
- e. the chief extensors of the leg

39. If the medial border of the foot is numb to the touch, which nerve is not functioning?

- A. tibial
- B. deep peroneal
- C. superficial peroneal
- D. saphenus
- E. medial sural cutaneous

40. If the foot is permanently dorsiflexed (extended) and everted, which nerve is NOT functioning properly?

- A. femoral
- B. tibial
- C. deep peroneal
- D. common peroneal
- E. superficial peroneal

41. The lumbrical muscles are located in which layer of the sole of the foot?

- A. first
- B. second
- C. third
- D. fourth
- E. fifth

42. Which of the following is found superficially immediately anterior to the medial malleolus?

- A. small saphenous vein
- B. greater saphenous vein
- C. dorsalis pedis artery
- D. arcuate artery
- E. anterior tibial artery

43. Which one of the following is the main invertor of the foot:

- A. peroneus brevis
- B. tibialis posterior
- C. flexor hallucis longus
- D. flexor digitorum longus
- E. peroneus longus

44. Following a car accident and some healing time, a patient is not able to rise from a sitting position. This could be due to a lesion of which of the following nerves?

- A. femoral
- B. obturator
- C. nerve to iliopsoas
- D. superior gluteal
- E. inferior gluteal

45. Which of the following dorsiflexes and inverts the foot?

- A. tibialis anterior
- B. extensor hallucis
- C. extensor digitorum longus
- D. tibialis posterior
- E. peroneus longus

46. The medial border of the femoral triangle is formed by the:

- A. sartorius muscle
- B. adductor longus muscle
- C. inguinal ligament
- D. pubic tubercle
- E. none of the above

47. The dorsalis pedis artery is usually a direct continuation of which artery?

- A. femoral
- B. popliteal
- C. peroneal
- D. anterior tibial
- E. posterior tibial

48. Following a football injury, a physician grabbed hold of a player's leg and noticed that the tibia could be moved anteriorly with freedom, especially when the knee was flexed. This disability should be diagnosed as a torn:

- A. anterior cruciate ligament
- B. lateral collateral ligament
- C. medial collateral ligament
- D. posterior cruciate ligament
- E. oblique popliteal ligament

49. A misplaced intramuscular injection into the gluteal region infiltrated the sciatic nerve and its surroundings, giving rise to temporary paralysis of the muscles supplied by that nerve. All of the following would be involved EXCEPT for the:

- A. tibialis anterior
- B. soleus
- C. peroneus brevis
- D. gracilis
- E. biceps femoris

50. Which of the following is NOT a lateral rotator of the hip?

- A. piriformis
- B. gluteus medius
- C. obturator internus
- D. obturator externus
- E. quadratus femoris

51. Paralysis of the quadriceps femoris results MAINLY in loss of:

- A. lateral rotation of the leg
- B. extension of the leg
- C. flexion of the thigh
- D. extension of the thigh
- E. flexion of the leg

52. Dislocation of the hip results in a vascular necrosis of the hip joint due to injuries to all of the following arteries, EXCEPT:

- a) Medial femoral circumflex artery
- b) Inferior gluteal artery
- c) Obturator artery
- d) Lateral femoral circumflex artery
- e) Deep iliac circumflex artery

3. Which of the following is **wrong** about knee joint:

- a. anterior cruciate ligament prevents backward displacement of femur on tibia
- b. muscles that lock the knee into full extension is the tensor fasciae latae and gluteus maximus by way of the iliotibial tract
- c. The knee is extended during the swing through phase and flexed during the push off phase
- d. Lateral meniscus is injured in Unhappy triad of the knee joint
- e. Prepatellar bursitis impairs flexion and extension of the knee joint

54. The largest branch of femoral artery within the femoral triangle is the:

- a. Superficial circumflex iliac artery
- b. Profunda(deep femoral) artery
- c. Superficial epigastric artery
- d. Superficial external pudendal artery
- e. Descending genicular artery

55. In a hunting accident, an arrow pierces the mid-calf of a hunter. A major artery is lacerated in the posterior leg, and you notice that the sole of his foot is cold and pale. The dorsum of the foot is warm and normally colored. The artery that seems to be injured is the:

- a. Fibular artery
- b. Posterior tibial artery
- c. Femoral artery
- d. Popliteal artery
- e. Medial plantar artery

56. The medial plantar nerve has motor branches to all of the following, **except**:

- a. Abductor hallucis
- b. Flexor digitorum brevis
- c. Adductor hallucis
- d. Flexor hallucis brevis
- e. Lumbrical - I

57. Which of the following nerves is paralyzed in "waddelling gait"?

- a. Superior gluteal
- b. Inferior gluteal
- c. femoral
- d. obturator
- e. sciatic

58. All of the following structures are injured in the "Unhappy Triad" of the knee joint, **EXCEPT**:

- A. anterior cruciate ligament
- B. lateral collateral ligament
- C. medial collateral ligament
- D. medial meniscus
- E. none of the above

59. Equinovarus or club-foot is the result of which one of the following nerves?

- A. femoral
- B. tibial
- C. deep peroneal
- D. common peroneal
- superficial peroneal

60. The following are the contents of the adductor canal, **except**:

- a. Saphenous nerve
- b. Femoral nerve
- c. Femoral artery
- d. Femoral vein
- e. Nerve to vastus medialis

Instruction: For each of the following multiple choice questions choose the one appropriate answer and write the corresponding letter on the attached answer sheet.

1. The wrong statement about the sternoclavicular joint is
 - a) It is a multiaxial joint
 - b) Its stability depends on the ligaments and the articular disc than bones and muscles
 - c) Its arterial supply is by internal thoracic artery and suprascapular artery
 - d) It's innervations is by medial supraclavicular nerve and nerve to subclavius muscle
 - e) None
2. The wrong association about cutaneous nerves of the upper limb their origin is
 - a) Lower lateral cutaneous nerve of the arm-----radial nerve
 - b) Lateral cutaneous nerve of the forearm-----musculocutaneous nerve
 - c) Upper lateral cutaneous nerve of the arm-----axillary nerve
 - d) Supraclavicular nerves-----cervical plexus
 - e) None
3. Incorrect association about anatomical defects and their location or causes is
 - a) Tennis elbow-----lateral epicondyle
 - b) Posterior interosseous nerve entrapment-----extensor digitorum
 - c) Triceps tendon reflex-----C7
 - d) Ulnar nerve paralysis-----medial epicondyle
 - e) None
- 4) Which of the following is wrong about the deep palmar arterial arch
 - a) It lies across the metacarpal bones, just distal to their branches
 - b) It is formed mainly by the radial artery
 - c) It is closer to the wrist than the superficial palmar arterial arch
 - d) It gives rise to palmar metacarpal arteries
 - e) None

105. Commissural fibers

- a. Connect corresponding areas of the right and left hemispheres
- b. Connect different parts of the same hemisphere
- c. Connect fibers entering and leaving the cerebral hemispheres from or to lower structures
- d. Largest in the corpus callosum
- e. A and D

106. Damage to the substantia nigra causes a decrease in the neurotransmitter dopamine gradually and generalized increase in muscle tone, which is the main symptom of ____

- a) Parkinson disease
- b) Rabies
- c) Myasthenia gravis
- d) Alzheimer disease
- e) Tetanus

107. Which one of the following pathways **doesn't** use thalamus as relay station?

- a. Olfactory pathway
- b. Trigeminal pathway
- c. Taste pathway
- d. visual pathway
- e. vestibular pathway

108. Which one of the following **does not** form circle of Willis?

- a. Anterior cerebral artery
- b. Posterior cerebral arteries
- c. Anterior communicating arteries
- d. Posterior communicating arteries
- e. Basilar artery

109. Which one of the following structures serves as reflex center for vision and hearing?

- a. Midbrain
- b. Medulla oblongata
- c. Cerebrum
- d. Pons
- e. Cerebellum

110. Two inferior colliculi of corpora quadrigemina that control reflex movements of the head, neck, and trunk in response to ____

- a. auditory stimuli
- b. gustatory stimuli
- c. visual stimuli
- d. olfactory stimuli
- e. all

111. Where does usually lumbar puncture take place?

- a. Between L1 and L2 vertebrae
- b. Between S1 and S2 vertebrae
- c. Between L2 and L3 vertebrae
- d. Between L5 and S1 vertebrae
- e. Between L3 and L4 vertebrae

112. Which one of the following are deep grooves generally dividing large regions or lobes of the brain?

- a. Gyri
- b. ridges
- c. sulci
- d. fissures
- e. folia

113. Which one of the following foramen is located between 3rd and 4th ventricle?

- a. Foramen of Monro or interventricular foramen
- b. Cerebral aque duct or foramen of Sylvian
- c. Foramen Magendie
- d. Foramen Luschka
- e. Lateral apertures

114. Which one of the following is true about the drainage of CSF?

- a. 3rd ventricle drains into lateral ventricles
- b. 4th ventricle drains into lateral ventricles
- c. Lateral ventricles drain into subarachnoid space / cistern of great cerebral vein
- d. 4th ventricle drains into subarachnoid space / cistern magna
- e. 4th ventricle drains into 3rd ventricle

115. Which one of the following statement is false about the role of CSF?

- a. Cushions and insulates delicate nervous tissue
- b. Exchange of plasma and formed elements
- c. Gives buoyancy to the brain and nourishes the brain
- d. Exchange of gases (O₂ and CO₂) , nutrients and wastes
- e. Forms as a filtrate of blood in choroid plexus and conducts chemicals signals between parts of CNS

116. Which one of the following regions of the brain contains paired structures which are main processing center for information and forms the lateral wall of the 3rd ventricle?

- a. Cerebrum b. diencephalon c. midbrain d. hindbrain e. brainstem

117. Which one of the following is true about the relations of thalamus?

- a. Ventrally subthalamus and hypothalamus
- b. Dorsally lateral ventricle
- c. Medially 3rd ventricle
- d. Laterally internal capsule
- e. all

118. Which one of the following is false about the red nucleus?

- a. Rostral to the decussation at the level of the superior colliculus the portion of the segmentum is occupied by red nucleus
- b. Some of the fibers of the superior cerebellar peduncles terminate in the red nucleus
- c. Degeneration of the pars compacta of the red nucleus is associated with Parkinson's diseases
- d. The red nucleus is involved in motor control
- e. The red nucleus projects to the inferior olivary nucleus of the medulla via the central segmental tract

119 Where does auditory pathway decussate or cross over?

- a. Red nucleus of red nucleus
- b. Trapezoid body of pons
- c. pyramid of medulla
- d. thalamus of diencephalon
- e. optic chiasma

120 If lesion is takes place in the spinal cord results in _____

- a. Ipsilateral paralysis
- b. Contralateral paralysis
- c. bilateral loss of sensation
- d. contralateral and bilateral paralysis
- e. none

GOOD LUCK!!!

DEPARTMENT OF ANATOMY
FACULTY OF MEDICINE
ADDIS ABABA UNIVERSITY
ANATOMY MAKE-UP AND REMEDIAL EXAMINATION FOR YEAR I M.D.
STUDENTS AC. YEAR 2010/11

Name: Rahel Getachew

Time allowed: two and half hours

Date: April 5th, 2011

Total No of pages= 17

1. Choose the best answer among the given alternatives.
(5 mark)

1. All of the following are cutaneous branches of the ventral rami of the upper four spinal nerves except:

- a. Greater auricular nerve
- ☒ b. Greater occipital nerve
- c. Lesser occipital nerve
- d. Transverse cervical nerve
- e. Supraclavicular nerve

2. All of the following happen at the level of C₆ vertebra except:

- a. Pharynx becomes esophagus ✓
- b. Larynx becomes trachea ✓
- c. Vertebral artery enters the transverse foramen ✓
- ☒ d. Vertebral vein exits the transverse foramen
- e. All of the above

A patient comes to the Out Patient Department (OPD) with a left trapezius tear. Which of the following symptoms might she also have?

- a. Left diaphragm paralysis
- ☒ b. Weakness in turning her head to the left
- c. Weakness in turning her head to the right
- d. Hoarse voice
- e. Difficulty swallowing

3. The platysma receives motor innervations from:

- a. Cranial nerve V
- ☒ b. Cranial nerve VII
- c. Ansa cervicalis
- d. Cranial nerve XI
- e. Cervical plexus

25. A 40-year-old woman presents at the out patient clinic with yellow pigmentation of the skin and sclera. Which of the following condition would most likely caused the obstructive jaundice?
- a. Aneurysm of the splenic artery
 - b. Perforated ulcer of the stomach
 - ☒ c. Cancer of head of pancreas
 - d. Damage to the pancreas during splenectomy
 - e. Cancer of body of pancreas
26. All of the following muscles possess attachment into linea alba **except**:
- a. External oblique abdominis
 - b. Internal oblique abdominis
 - c. Transverse abdominis
 - d. pyramidalis
 - ☒ e. none
27. Anastomosis of the left gastric artery with the right gastric artery occurs:
- ☒ a. on lesser curvature of the stomach
 - b. on greater curvature of the stomach
 - c. on inferior border of the liver
 - d. on inferior border of liver
 - e. none
28. An eighteen years old girl admitted to Black Lion Hospital yesterday was found to have infection of her vagina below her hymen. Diagnosis revealed that metastasis had occurred primarily to other organs. As a result which lymph nodes are may be enlarged?
- a. external iliac
 - b. internal iliac
 - c. superficial inguinal
 - d. deep inguinal
 - e. all of the above
29. A radiological examination of a patient revealed a large tumor in the quadrate lobe of the liver. During the surgical removal of the tumor, one of the vessels that needs to be clamped to effectively control bleeding is the
- a. Left hepatic artery
 - b. Right hepatic artery
 - ☒ c. Cystic artery
 - d. Splenic artery
 - e. Right gastric artery
30. If abdominal infection spreads retroperitoneally, which of the following structures would most likely to be affected?
- a. stomach
 - b. jejunum
 - c. spleen
 - ☒ d. Adrenal gland
 - e. none of the above

Following an emergency appendectomy your patient complained of having paresthesia (numbness) of the skin at the pubic region. The most likely nerve that has been injured during the operation is:

- a. Genitofemoral
- b. Subcostal
- ☒ c. Iliohypogastric
- d. Spinal nerve T10
- e. Spinal nerve T9

32. Which of the following characteristics are not typical of the large intestine?

- a. Teniae coli
- b. Sacculations
- c. Haustrations
- d. Appendicis epiploicae
- ☒ e. none

33. Visceral peritoneum is sensitive to pain because it is innervated by:

- a. SNS
- ☒ b. ANS
- c. ANS and SNS
- ☒ d. None
- e. All

34. Posterior relation of the stomach is formed by all except:

- a. Pancreas
- b. gall bladder
- c. spleen
- d. left kidney
- e. left suprarenal

35. All of the following statements contrasting male and female pelvic structures are true except

- a. In females, the pelvic inlet is oval, in males it is more heart shaped.
- b. The tip of coccyx is more inverted in females than males
- ☒ c. The sacrum is wider in females than in males.
- d. All pelvic diameters are greater in females than males
- e. The subpubic angle is greater in females than males

36. Muscles of urogenital diaphragm includes

- ☒ a. Levator ani and coccygeus
- b. Bulbospongiosus and ischiocavernosus
- c. Ani externus and internus
- d. Deep transverse perineal and sphincter urethrae
- e. None of the above

37. Part of broad ligament attached to the ovary is

- a. mesometrium
- ☒ b. mesovarium
- c. mesoappendix
- d. mesocolon
- e. none of the above

- 0911315622
- The normal position of uterus is
- retroflexed and anteverted
 - anteverted and retroflexed
 - anteverted and anteverted
 - retroverted and retroflexed
 - anteverted and retroverted
9. Vas deferens continuous with
- tail of epididymis
 - head of epididymis
 - body of epididymis
 - all of the above
 - none of the above
10. Wrong about Ovary is
- Weights about 15 grams
 - Innervated by pelvic splanchnic nerves
 - Almond shaped
 - Measures 3 cm
 - Drains into renal vein on left side
41. Perineal muscle which helps to maintain erection by compressing the veins in the crus is
- bulbospongiosus
 - superficial transverse perineal muscles
 - deep transverse perineal muscles
 - ischiocavernosus
 - sphincter urethrae
42. Bulbourethral glands in male are homologous to ----- in female.
- Greater vestibular glands
 - Prostate
 - Seminal vesicle
 - All
 - none
43. Most of the ducts of prostate glands open into
- prostatic utricle
 - prostatic sinus
 - seminal colliculus
 - verumontanum
 - all of the above
44. All of the following muscles are paralyzed if pudendal nerve is injured in the perineal canal except:
- bulbospongiosus
 - ischiocavernosus
 - levator ani
 - internal anal sphincter
 - external anal sphincter
- JCF
FCS
VSC

Bulbourethral glands in male are homologous to _____ in female.

- a. Greater vestibular glands
 - b. Prostate
 - c. Seminal vesicle
 - d. All
 - e. none
3. Most of the ducts of prostate glands open into
- a. prostatic utricle
 - b. prostatic sinus
 - c. seminal colliculus
 - d. verumontanum
 - e. all of the above
14. All of the following muscles are paralyzed if pudendal nerve is injured in the pudendal canal **except**:
- a. bulbospongiosus
 - b. ischiocavernosus
 - c. levator ani
 - d. internal anal sphincter
 - e. external anal sphincter

Histology & Embryology Mid-Year Examination for Year PCI Medical Students

Date: August 04, 2014

Time allowed: 2:00 hours.

Name: _____

HISTOLOGY

For each of the following Multiple Choice Questions choose the one best appropriate answer and mark "X" in the middle of the appropriate box located against the corresponding question number and letter in the separate answer sheet provided.
(50 points, 1 point for each correct answer)

1. Which of the following is the function of ocular lens of light microscope?
☒ a) Magnification b) Resolution c) Focusing light to the specimen
d) a & b e) All of the above
2. Following Haematoxylin & Eosin (H & E) staining, the eosinophilic components of the tissue are:
☒ a) Stain red-pink b) Include the nucleus
c) May show metachromatic reaction d) Stain the ribosomes
e) Are acidic components of the tissue
3. Regarding to the nucleolus:
a) Is synonymous of the nucleus.
b) Is the site of mitochondrial synthesis.
c) Is membrane bound region within the nucleus.
d) Is made by regions of chromatins contributed by different chromosomes.
☒ e) Is not visible under the light microscope.
4. Organelles involved in the storage and release of calcium in the skeletal muscle are:
a) Rough endoplasmic reticulum ☒ b) Smooth endoplasmic reticulum
c) Peroxisomes d) Golgi bodies e) Microtubules

OLOGY

ice Questions, choose the one best
your choice

45. The alar plate of the neural tube gives rise to:
a) Motor neurons
b) Sensory neurons
c) Both motor and sensory neurons
d) Special visceral efferent neurones
e) None of the above.
46. The outer layer of optic cup gives rise to:
a) Ciliary muscles
b) Bipolar neurons of retina
c) Ganglion cells of retina
d) Pigment epithelial cells of retina
e) Choroid
47. Lens develops from:
a) Optic cup
b) Surface ectoderm
c) Diencephalon
d) Mesencephalon
e) Local mesenchyme
48. Cornea is derivative of:
a) Ectoderm
b) Mesoderm
c) Endoderm
d) a & b
e) All
49. Outer aspect of the tympanic membrane is developed from which germ layer?
a) Ectoderm
b) Mesoderm
c) Endoderm
d) a & b
e) All of the above
50. Membranous labyrinth of the inner ear develops from:
a) Diencephalon
b) Surface ectoderm
c) First pharyngeal pouch
d) Local mesenchyme
e) Mesencephalon

42. With regard to the cells of parathyroid gland:
 a) Chief cells are larger in size than the oxyphil cells.
 b) Oxyphil cells are more in number than the chief cells.
 c) Oxyphil cells flattened endothelial like cells.
 d) Chief cells occur as follicles.
 e) Oxyphil cells are more eosinophilic than the chief cells.
43. Nissl bodies of neurons represent:
 a) Golgi bodies
 b) Lysosomes
 c) Microfilaments
 d) Rough endoplasmic reticulum
 e) Smooth endoplasmic reticulum
44. Which type of neurones of the cerebral cortex is predominantly found in the motor areas?
 a) Cells of Martinotti.
 b) Horizontal cells of Cajal.
 c) Pyramidal cells.
 d) Granular cells.
 e) Fusiform cells.
45. Which layer of the cerebellar cortex is made by a synaptic field where all the five types of neurons of the cortex contribute to the synapse?
 a) Purkinje cell layer.
 b) Multiform layer.
 c) Molecular layer.
 d) Granular layer.
 e) None of the above.
46. The parallel fibres found in the molecular layer of the cerebellar cortex are made by axons of:
 a) Purkinje cells.
 b) Stellate cells.
 c) Basket cells.
 d) Granule cells.
 e) Golgi cells
47. Which of the following cells does not participate in the synapse formation at the outer plexiform layer of the retina as seen under the light microscope?
 a) Rods
 b) Cones
 c) Ganglion cells
 d) Horizontal cells
 e) Bipolar cells
48. Lacrimal gland is an example of:
 a) Mucouse producing gland
 b) Serous producing gland
 c) Mixed mucoserous gland
 d) Steroid producing gland
 e) None of the above
49. The transmission of sound vibrations through the internal ear occurs chiefly through:
 a) Nerve fibers
 b) Air
 c) Fluid
 d) Bone.
 e) All of the above
50. Ceruminous glands found in the external acoustic meatus are modified:
 a) Sebaceous glands
 b) Merocrine sweat glands
 c) Apocrine sweat glands
 d) Serous glands
 e) Mucous glands

- b 5. The plasma membrane:
- Has peripheral proteins which are mostly found loosely attached on the outer surface of the membrane.
 - Has integral proteins which show free lateral mobility and rearrangements.
 - Has oligosaccharides which are found projecting from both outer and inner surfaces.
 - Is involved in signal transduction through different classes of polysaccharides receptors.
 - Is about 7.5-10 μ m thick clearly visible under light microscopy
- e 6. Which of the following is not a function of the epithelial tissue?
- Absorption
 - Sensation
 - Secretion as glands
 - Contractility
 - None of the above
- e 7. Glandular cells which release their secretory product by exocytosis are known as:
- Holocrine
 - Monocrine
 - Polycrine
 - Apocrine
 - Merocrine
- d 8. Exocrine glands whose secretory end pieces are organized to resemble branched grapes and empty onto the covering epithelium by a single unbranched duct are:
- Simple branched tubular
 - Compound tubular
 - Compound branched tubular
 - Simple branched acinar
 - Compound acinar
- e 9. Which of the following is not the nature of ground substance of connective tissue?
- Act as mechanical support
 - Facilitate migration of cells
 - Facilitate rapid diffusion of water-insoluble molecules
 - Act as barrier for micro-organisms
 - None of the above
- e 10. What type of tissue is a tendon composed of?
- Mucous connective tissue
 - Mesenchyme
 - Loose irregular connective tissue
 - Dense irregular connective tissue
 - Dense regular connective tissue
- c 11. Cells responsible for synthesis of the fibres and ground substance of connective tissue are:
- Macrophages
 - Mast cells
 - Fibroblasts
 - Plasma cells
 - Histiocytes
- a 12. Which type of bone cells are involved in bone decalcification?
- Osteoclasts
 - Osteoblasts
 - Osteocytes
 - a & b
 - All
- e 13. The outer surface of a bone is lined by:
- Perichondrium
 - Endosteum
 - Endochondrium
 - Basal lamina
 - Periosteum
- e 14. The sarcomer of a skeletal muscle is the region between two
- A bands
 - I bands
 - H bands
 - Z lines
 - M lines

24. Which type of the reticuloepithelial cells of thymus makes Hassal's corpuscles?
 a) Type I b) Type II c) Type III d) Type IV e) Type V
25. Which of the following lymphoid organs is involved in filtering lymph?
 a) Appendix b) Palatine tonsil c) Spleen d) Thymus e) Lymph node
26. Which of the following statements is **not true** about the lip?
 a) The inner aspect has salivary glands
 c) The transitional border has no glands
 d) The inner surface is lined by stratified columnar epithelium
 e) All of the above
27. Which of the following statements is true regarding the parotid glands?
 a) Have no striated ducts.
 b) Have centroacinar cells at the centre of the acini.
 c) Are purely serous glands in the adult
 d) Are mixed mucoserous glands, but mainly serous.
 e) Are mixed mucoserous glands, but mainly mucous.
28. Which region of the stomach has long and wide pits with short glands?
 a) Cardiac b) Pyloric c) Body d) Fundus e) c & d
29. Which one of the following marks best the duodinum from the rest of small intestinal region?
 a) Presence of long and finger like villi
 c) Brunner's glands d) Peyer's patches b) Absence of tenia coli
 e) Crypts of Lieberkun
30. Which type of cells of gastric glands are responsible for the secretion of HCl?
 a) Chief cells b) Paneth's cells c) Parietal cells
 d) Mucous neck cells e) Surface mucous cells
31. Which of the following statements is true regarding the intestine?
 a) The striated border of the epithelial lining of the small intestine is made by microvilli.
 b) Intestinal glands in the large intestine are branched alveolar, mainly serous producing in type.
 c) Small intestine has teniae coli.
 d) Large intestine has villi.
 e) Muscularis externa of rectum is made by skeletal muscles.
32. Which of the following is **not found** in the appendix?
 a) Small and irregular lumen. b) Muscularis externa. c) Intestinal glands of Lieberkühn.
 d) Teniae coli. e) Lymphoid nodules in the lamina propria and submucosa.

EMBRYOLOGY

For each of the following Multiple Choice Questions, choose the one best appropriate answer and write the letter of your choice in the space provided in front of the question number.

(50 points, 1 point for each correct answer)

d 1. Site of normal fertilization is:

- a) Uterine cavity
- b) Vagina
- c) Cervix
- d) Ampullary region of the uterine tube
- e) Intramural portion of the uterine tube

b 2. Which statement is false about fertilization?

- a) It restores the diploid state of the chromosomes.
- b) It determines both the genetic and phenotypic sex of the zygote.
- c) Cause completion of the second mitotic division of the oocyte.
- d) Initiates the first cleavage division of the zygote.
- e) Causes metabolic activation of the oocyte.

a 3. The yolk sac contributes to the development of all of the following, except:

- a) Central nervous system
- b) Germ cells.
- c) Midgut.
- d) Hindgut.
- e) Allantois.

e 4. Which one of the following structures does not change its position because of the head fold?

- a) Oropharyngeal membrane
- b) Septum transversum
- c) Primitive heart
- d) All of the above
- e) None of the above

e 5. Which of the following contributes to the formation of the amniotic fluid?

- a) Mater maternal interstitial tissue fluid
- b) Blood in the intervillous spaces of placenta
- c) Fetal interstitial tissue fluid
- d) a & b
- e) All

d 6. Which of the following is not derivative of the septum transversum?

- a) Stroma of the liver
- b) Central tendon of the diaphragm
- c) Lesser omentum
- d) Splenorenal ligament
- e) Falciform ligament

b 7. Most birth defects occur during which of the following periods/stages of the intrauterine life?

- a) First 2 weeks
- b) Embryonic period
- c) Fetal period
- d) Last trimester
- e) Zygote

a 8. Fetal portion of placenta is formed by:

- a) Chorion frondosum
- b) Chorion laevae
- c) Smooth chorion frondosum
- d) Decidua basalis
- e) None of the above

EMBRYOLOGY

Choice Questions. choose the one best answer in the space provided in

- d 9. The cranial half of the 1st cervical sclerotome gives rise to:
 a) Upper part of body of the 1st cervical vertebra.
 b) Lower part of body of the 1st cervical vertebra.
 c) Whole body of the 1st cervical vertebra.
d) Part of base of occipital bone.
 e) Part of roof of the cranial cavity
- e 10. Which of the following muscles develops from the epimers?
 a) Intercostal muscles. b) The strap muscles of the neck.
 c) Rectus abdominis. d) Quadratus lumborum. e) Extensor muscles of the back.
- c 11. Failure fusion of the neural arches could lead to:
 a) Achondroplasia b) Acromegally c) Spina bifida
 d) Brachycephaly e) Amelia
- c 12. Which of the following gives rise to most muscles of the tongue?
 a) Lateral mesoderm b) Local splanchnic mesenchyme
c) Myotomes of the occipital somites d) All of the above e) None of the above
- a 13. Frontonasal prominence gives rise to the
 a) Forehead b) Sides of the face c) Sides of the nose
 d) All of the above e) None of the above
- e 14. Regarding to the development of the musculoskeletal system
 a) At 7th week upper limbs rotate 90° medially
 b) Derivatives of the epimers are innervated by the dorsal branches of the ventral primary rami of the spinal nerves.
 c) Muscles of the limb are derivative of intermediate mesoderm.
 d) Lower limb buds appear before those of the upper limb.
e) Skeletons of the upper limb develop from the somatic layer of the lateral mesoderm.
- b 15. In the craniofacial region
 a) The intermaxillary segment is formed by fusion of the two lateral nasal prominences.
b) The nasolacrimal groove separates maxillary process from the lateral nasal prominence.
 c) The secondary palate is formed by the fusion of the palatal shelves that develop from underneath of the frontonasal processes.
 d) The upper jaw develops in the 2nd pharyngeal arch cartilages.
 e) The nasal septum develops as upward projection from the maxillary process.
- c 16. The ribs develop from:
 a) Somatic layer of lateral mesoderm. b) Visceral layer of lateral mesoderm.
c) Paraxial mesoderm. d) Intermediate mesoderm. e) Neural crest.
- d 17. Which of the following bones begins to ossify before the others?
 a) Vertebra b) Scapula c) Humerus
d) Clavicle e) Femur

RYOLOGY

Multiple Choice Questions. Choose the one best answer. Write the letter of your choice in the space provided in the margin. 1 point for each correct answer.

- d 26. Primitive vein that gives rise to superior vena cava is:
 a) Right supracardinal vein
 b) Right subcardinal vein
 c) Right vitelline vein
 d) Right common cardinal vein
 e) Right posterior cardinal vein
- e 27. In fetal blood circulation oxygenated blood is **not** contaminated by deoxygenated blood in which of the following sites?
 a) Left atrium
 b) Right atrium
 c) Inferior vena cava
 d) Descending aorta
 e) None of the above
- e 28. The embryonic foregut gives rise to all of the above except:
 a) Liver
 b) Pancreas
 c) Lung
 d) Esophagus
 e) None of the above
- c 29. The junction between derivatives of stomodaeum and foregut is located at:
 a) Oropharyngeal arch
 b) Middle of pharynx
 c) Just behind the alveolar arches
 d) Oropharyngeal Junction
 e) None of the above
- b 30. What adult structure is formed from the caudal limb of the primary intestinal loop?
 a) Jejunum
 b) Cecum
 c) Descending colon
 d) Sigmoid colon
 e) Duodenum
- a 31. The terminal dilated part of the hindgut is called the:
 a) Cloaca
 b) Yolk stalk
 c) Allantois
 d) Cecum
 e) Coelom
- c 32. A fetus born prematurely during which of the following period of lung development could survive?
 a) Canalicular
 b) Pseudoglandular
 c) Terminal sac period
 d) None of the above
- c 33. Which of the following cells of the lung develops from mesoderm?
 a) Type I Pneumocytes.
 b) Type II Pneumocytes.
 c) Alveolar dust cells.
 d) Clara cells
 e) Goblet cells of trachea
- c 34. Which of the following is **not** a site for blood cell formation during intrauterine life?
 a) Spleen
 b) Lymph node
 c) Kidneys
 d) Liver
 e) Allantois
- e 35. Thymus develops mainly from:
 a) 2nd pharyngeal cleft
 b) 3rd pharyngeal cleft
 c) 4th pharyngeal cleft
 d) 2nd Pharyngeal pouch
 e) 3rd pharyngeal pouch

7) Which of the following is **NOT** a function of the kidney?

- a) Erythropoietin production
- b) Vitamin D modification
- c) Acid base balance
- d) Aldosterone production
- e) Renin production

8) Which of the following is **NOT** the function of mesangial cells?

- a) Structural support to the glomerulus.
- b) Synthesis of extracellular matrix.
- c) Endocytosis and contractility.
- d) Maintenance of acid-base balance in the blood.
- e) Increase blood flow and the effective surface area available for filtration.

9) Which of the following is **NOT** considered part of the male genital duct system?

- a) Rete testis
- b) Tubuli recti
- c) Seminal vesicle
- d) Ductus deferens
- e) Ductus epididymidis

10) Select the **incorrect** statement in the male reproductive system.

- a) During arousal, the testes normally descend into the scrotum, otherwise they are abdominal organs.
- b) The bulbourethral glands in males are homologous to the Bartholin's glands in females.
- c) The prostate gland is made of compound tubuloalveolar glands.
- d) The glands of the prostate lie in a fibromuscular stroma.
- e) The prostate gland is the largest accessory structure of the male reproductive system.

11) Testis has the following coverings *except*

- a) Tunica albuginea
- b) Tunica vasculosa
- c) Tunica albuginea
- d) Tunica vaginalis
- e) All of them

12) Fructose, a potential fuel for mitochondria of spermatozoa, is found in the highest concentrations in secretions of which of the following glands?

- a) Bulbourethral gland
- b) Glands of Littre
- c) Prostate gland
- d) Seminal vesicle
- e) Testis

13) **Spermiogenesis**

- a) Occurs before puberty.
- b) Involves cytodifferentiation of early spermatids.
- c) Involves spermatocytogenesis, meiosis and spermatogenesis.
- d) Takes place in diploid cells.
- e) Results in the production of primary spermatocytes.

14) In the male reproductive system, which statement is **false**?

- a) Seminal vesicles are not reservoirs for spermatozoa.
- b) 70% of human ejaculate originates in the prostate gland.
- c) Height of epithelial cells of the seminal vesicle and the degree of activity of the secretory processes are dependant of testosterone levels.
- d) The tubuloalveolar glands of the bulbourethral glands are formed by simple cuboidal epithelium.
- e) The number of corpora amylacea increases with age.

15) Which statement is **false** about the Sertoli cells?

- a) Their bases adhere to the basal lamina.
- b) They produce testicular androgens.
- c) They may be connected by gap junctions that provide ionic and chemical coupling of the spermatogenic cells.
- d) They do not divide during the reproductive period.
- e) They are extremely resistant to adverse conditions such as infections, malnutrition.

- 16) Concerning the male reproductive system, which statement is false?
- a) During spermiogenesis, the spermatids quickly undergo a second meiotic division to give spermatozoa.
 - b) The ductus epididymidis is lined by pseudostratified columnar epithelium.
 - c) Sertoli cells bear the primary responsibility for forming the blood testis barrier.
 - d) The seminiferous tubules are lined with a complex stratified epithelium.
 - e) The primary spermatocytes are the largest cells of the spermatogenic lineage.
- 17) Which of the following is true of the uterine tube's ampulla but not of its isthmus?
- a) Contains mucus-secreting peg cells
 - b) Contains ciliated epithelial cells
 - c) Is covered by serosa derived from the peritoneum
 - d) Contains smooth muscle in its wall
 - e) Contains large mucosal folds that are extensively branched
- 18) Which of the following is true of the stratum basale but not of the stratum functionalis of endometrium?
- a) Includes the uterine surface epithelium.
 - b) Includes connective tissue
 - c) Contains cells that are highly responsive to ovarian hormones.
 - d) Is a layer that undergoes cyclic thickening and shedding
 - e) Contains cells that replace the surface epithelium after menstruation.
- 19) Concerning the female reproductive system, which of the following statements is false?
- a) During pregnancy hCG maintains the corpus luteum.
 - b) Ovulation is triggered by a dramatic increase in LH.
 - c) Primordial follicle is arrested in prophase stage.
 - d) The life span of corpus luteum of menstruation is 10 lunar months.
 - e) The superficial cells of the stratified squamous epithelium of vagina synthesize and store glycogen.
- 20) Which of the following statements is false in the female reproductive system?
- a) Nonciliated secretory cells present in the lining epithelium of uterine tube are called peg cells.
 - b) Mammary gland is a modified apocrine sweat gland.
 - c) Granulosa cells that are adherent to the zona pellucida form corona radiata.
 - d) Granulosa lutein cells of corpus luteum secrete chorionic gonadotropin.
 - e) Fornix of vagina is lined by stratified squamous epithelium.
- 21) Which of the following statements is false in the female reproductive system?
- a) Primordial follicle is identified in section by the presence of single layer of flattened follicular cells.
 - b) Uterine tube is lined by simple columnar ciliated epithelium.
 - c) Primary oocyte completes its first meiotic division at the time of ovulation.
 - d) Proliferative phase of uterus coincides with the development of ovarian follicles.
 - e) None of the above.
- 22) Which of the following statements is true in the female reproductive system?
- a) Germinal (surface) epithelium covering the ovary gives rise to ova.
 - b) Glassy membrane is the last structure to disappear in an atretic follicle.
 - c) The alkaline environment of vagina prevents growth of microorganisms.
 - d) The cells of theca interna secrete progesterone.
 - e) Zona pellucida is derived from liquor folliculi.

Part II:

Embryology

For each of the following Multiple Choice Questions, choose the one best appropriate and mark "X" in the middle of the appropriate box located against the corresponding number and letter in the separate answer sheet provided. (40 points, 1 point correct answer)

- When monozygotic twins are formed by split at morula stage, the twins will have
 - Same chorionic sac and placenta, but separate amniotic cavity
 - Same chorionic sac, placenta and amniotic cavity
 - ☒ Separate chorionic sac, placenta and amniotic cavity
 - Separate chorionic sac and placenta, but same amniotic cavity
- Conjoined twins are most frequently formed when there is an incomplete split at which of the following stages of development?
 - Two cell stage
 - Morula stage
 - Early blastocyst stage
 - ☒ Bilaminar germ disc
- Congenital anomalies refer to which types of the following abnormalities present at birth?
 - Structural disorders
 - Functional disorders
 - Behavioural disorders
 - ☒ a & b
 - ☒ All of the above
- Frequencies of occurrence of major developmental defects in early embryos as compared to the newborn respectively:
 - 2-3% and 10-15%
 - ☒ 10-15% and 2-3%
 - 30% and 40-50%
 - 50% and 30%-40%
 - Same at both stages as 10%
- Which of the following organs is most affected by human congenital anomalies present at birth?
 - ☒ Brain
 - ☒ Heart
 - Kidneys
 - Limbs
 - All have equal chance showing no difference
- With regard to the congenital anomalies, which of the following statements is false?
 - ☒ At birth, the incidence of minor congenital anomalies is less than that of major congenital anomalies.
 - Infants with multiple minor anomalies most frequently have one or more associated major anomalies.
 - Recessively inheritable mutations may cause birth defect while occurring in single (heterozygous) dose.
 - Monosomy of the sex, but not of the autosomal chromosomes are known to cause human birth defect.
 - A similar teratogen may not cause identical congenital anomalies in different genomes.
- Congenital anomalies that are caused because of mechanical forces that mould a part of the fetus over a period of time are referred to as:
 - Malformations
 - Disruptions
 - ☒ Deformations
 - Dysplasia
 - Syndrome
- The causes for most of the currently detected major human birth defects at birth are:
 - Genetic factors
 - Environmental factors
 - Combination of genetic and environmental factors
 - ☒ Unknown factors
- A condition where there is a decrease or an increase from the normal diploid number of a specific chromosome cause(s) for birth defect is known as: